how much grass is returned to a healthy grassland, in the form of residual biomass, in northern Ghana, in tons per hectare? What about a degraded grassland?

For northern Ghanaian Guinea savanna grasslands, total standing herbaceous biomass at the end of the growing season is typically on the order of 3–8 t dry matter ha^{-1} in relatively intact, well-managed sites, of which perhaps 1–4 t ha^{-1} remains as residual biomass after grazing and early burning; heavily degraded, overgrazed grasslands often have ≤1–2 t ha^{-1} total standing biomass and <1 t ha^{-1} residual.[1]

Healthy grassland (Guinea savanna, northern Ghana)

- IPCC default values for West African Guinea-zone savannas give maximum aboveground herbaceous biomass around 4–8 t dry matter ha^{-1}.
- Regional savanna syntheses underlying these defaults describe growing-season average biomass in tropical African savannas around 6.6 t ha^{-1}, with Guinea savannas toward the upper part of the 3–8 t ha^{-1} range.
- Experimental and grazing-study data from the Guinea savanna zone of Ghana show substantial herbaceous biomass production and moderate stocking rates, consistent with several tonnes per hectare of grass at peak standing crop, implying on the order of 1–4 t ha^{-1} of ungrazed/residual material left on the ground in well-managed or lightly grazed pasture at the end of the season.[2]

Given those constraints, a reasonable working range for residual biomass in a **healthy** northern Ghana grassland is about **1–4 t dry matter ha^{-1} yr^{-1}**, with 2–3 t ha^{-1} as a central, methodologically conservative value for SOC modeling.[1]

Degraded grassland (overgrazed / frequently burned)

- The same IPCC table distinguishes more arid Sahel/North Sudan savannas with much lower maximum herbaceous biomass (often 0.5–4 t ha^{-1}), reflecting how degradation and aridity reduce fuel loads.
- The guidance explicitly notes that frequent burning and overgrazing can lead to long-term degradation, reducing aboveground biomass and soil carbon, i.e., less regrowth and less fuel per hectare over time.
- Reviews of degraded West African savannas report strongly reduced grass cover and productivity, often down to ≤1–2 t ha^{-1} total standing herbaceous biomass under chronic overgrazing and frequent fire.[3]

For a **degraded** northern Ghana grassland, it is therefore defensible to assume **≤1–2 t dry matter ha⁻¹ yr⁻¹** total standing biomass at the end of the season, with **residual biomass often <1 t ha⁻¹**, and a central value around 0.5–1 t ha⁻¹ if you need a single conservative number for residual inputs. [3]

Practical ranges you could use

If you need explicit numbers for modeling (e.g., RothC inputs):

- Healthy Guinea-savanna grassland, northern Ghana:
  - Peak aboveground herbaceous biomass: 4–8 t DM ha⁻¹. [1]
  - Residual biomass returned (litter + ungrazed + trampled): ~1–4 t DM ha⁻¹ yr⁻¹ (use 2–3 t ha⁻¹ as a central value). [2]
- Degraded, overgrazed/burned grassland:
  - Peak aboveground herbaceous biomass: ~0.5–2 t DM ha⁻¹. [3]
  - Residual biomass returned: typically <1 t DM ha⁻¹ yr⁻¹ (0.5–1 t ha⁻¹ as a central value). [3]

If you tell me your specific rainfall band and grazing/burning regime, I can narrow those ranges further and translate them directly into annual C-input values.

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Would I multiply these numbers by 0.5, for calculating the C-input parameter in RothC?

Yes, multiplying by about 0.5 is a reasonable and commonly used way to convert dry biomass (DM) to carbon input for RothC, because plant dry matter is typically around 45–50% carbon by mass. [13] [14]

How to convert to RothC C-input

1. Start from your estimated residual biomass in t DM ha⁻¹ yr⁻¹ (e.g., 2 t DM ha⁻¹ yr⁻¹ for healthy, 0.7 t DM ha⁻¹ yr⁻¹ for degraded). [15]
2. Multiply by a carbon fraction of 0.45–0.5 to get t C ha⁻¹ yr⁻¹: many RothC applications and related tools assume about 0.45–0.5 C fraction of plant residues. [14] [13]
3. Use that result as the annual plant C input parameter in RothC (before splitting into DPM/RPM, which RothC does internally based on the DPM/RPM ratio). [16] [17] [18]

Example

- Healthy grassland residual biomass: 2 t DM ha⁻¹ yr⁻¹ → $2 \times 0.5 = 1$ t C ha⁻¹ yr⁻¹ C-input. [14]
- Degraded grassland residual biomass: 0.7 t DM ha⁻¹ yr⁻¹ → $0.7 \times 0.5 \approx 0.35$ t C ha⁻¹ yr⁻¹ C-input. [14]

If you want to be more conservative for SOC credits, you could use 0.45 instead of 0.5 as the C fraction; the choice just needs to be documented consistently. [13] [14]

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What about the amount of manure deposited to a regular grassland in northern Ghana, in tons per hectare, roughly?

For a “regular” (moderately stocked, not heavily degraded) grassland in northern Ghana, a rough, order-of-magnitude estimate is on the order of **1–5 t fresh manure ha^{-1} yr^{-1} actually dropped on the pasture**, with a lot of spatial concentration near kraals/bomas and water points. [24] [25] [26]

How to back-out that range

- Guinea-savanna grazing-capacity mapping in Ghana reports grazing capacities from about 0.18 to 3.79 livestock units (LU) ha^{-1}, spanning very lightly to heavily stocked sites. [25]
- USDA/extension-type data suggest one 450 kg bovine produces roughly 25–60 kg fresh manure per day, or about 9–22 t fresh manure per head per year, depending on diet and system. [27] [28] [29]
- For a more realistic “regular” savanna stocking rate of perhaps 0.2–0.7 LU ha^{-1} in northern Ghana (mid-range of the Ghana grazing-capacity study, excluding the most heavily stocked sites), this implies something like 2–15 t fresh manure ha^{-1} yr^{-1} if cattle spent all their time on that hectare. [26] [25]
- In practice, herds move and spend only part of the day actually grazing a given paddock, and much manure accumulates in night enclosures, so only a fraction of total excretion ends up on any given grazed hectare; using something like 10–30% of that gross theoretical deposition leads to about **1–5 t fresh manure ha^{-1} yr^{-1}** as a plausible working range for manure actually deposited per hectare of grazed pasture. [24] [25]

Using this for SOC / RothC

- If you treat manure separately from plant residues in your C-input bookkeeping, you can assume on the order of 1–5 t fresh manure ha^{-1} yr^{-1} to pasture, which at roughly 20–25% dry matter corresponds to ~0.2–1.25 t manure DM ha^{-1} yr^{-1}. [28] [29]
- Using a manure C fraction of ~0.4–0.45 of DM, that is roughly 0.08–0.56 t C ha^{-1} yr^{-1} of additional manure-derived C input to soil, on top of the plant-residue inputs you already estimated. [30] [31] [28]

If you tell me your assumed stocking rate (LU ha^{-1}) and how many hours per day animals are actually on the pasture versus in pens, I can tighten those numbers and give you a per-hectare C-input from manure directly.

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